

Local and Global Sequence Alignments

- Dot Plot does not resolve similarity that is interrupted by regions that do not match very well or that are present in only one of the sequences
- ~~Align~~ Alignment Column is a match between two sequences of interest
- A Gap is an arbitrary unit character that is represented by a dash
- ~~Align~~ Alignment cannot tell us whether origin of a gap (probable)
- "Sequence Identity" for this column, the nucleotides from the sequences are identical
 - ↳ Check site of a protein or a gene
- Similarity: for this column, nucleotides from the sequences are identical, but not necessarily similar chemical properties like G and A being purines. (also pyrimidines)
- To align two sequences of 300 nucleotide lengths, it takes 10^8 comparisons
- Needleman and Wunsch in 1970 proposed that the solution is to break the problem down into a progressive building by comparing two residues at a time
 - ↳ Called dynamic programming
 - ↳ Step 1: Generate all possible (trial) alignments of the two sequences
 - ↳ Step 2: Use a scoring system to score the trial alignments to find the best one
- Match = 1, Mismatch = 0, Gap = -1, highest is the optimal alignment
- Smith and Waterman recognized that the most biologically significant regions in DNA and protein sequences were sub-regions that align very well.
- Local Alignment are more meaningful than global alignment because they conserve patterns
- Local Alignment (het end to end), take the steps of Needleman-Wunsch
- Needleman takes for ~~Alignment~~ Alignment
- Smith Waterman is a modified version of Needleman-Wunsch algorithm (given different scoring and termination)
- Either one can be local or global depending on scoring